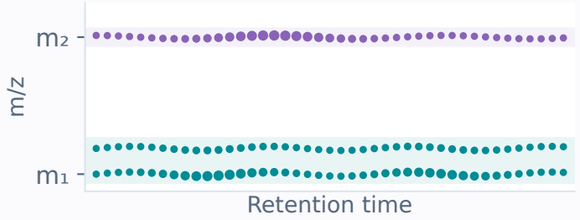
The difference is how peak candidates are formed

Both approaches start with retention time, m/z and intensity. They organise the data differently.

TRACE-BASED DETECTION

1 Assemble traces / ROIs



2 Find peaks within traces



3 Report peak candidates

A trace is the parent object. Chromatographic segmentation defines peaks inside it.

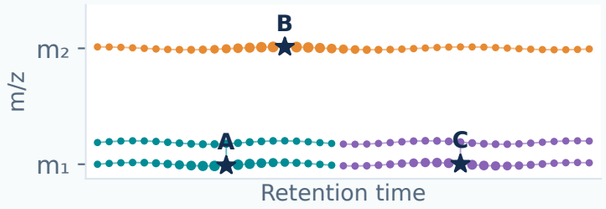
Trace building may use intensity and adaptive tolerances; implementations differ.

Peak3D CANDIDATE DETECTION

1 Link local centroids uphill



2 Form and merge apex basins



3 Validate peak candidates

A basin is the candidate object. Apply basin gates, an EIC check, and the final score.

Dot size shows intensity. Peak3D panels show actual graph and basin assignments.

Trace-based: form traces, then find peaks. Peak3D: form apex basins directly from centroids.

Architecture schematic • centWave / MassCube: ROI then segmentation; asari: composite mass tracks • sources in notes